

Raman Saparkhan

☎ +974 5037 7425 | ✉ rsaparkh@andrew.cmu.edu | [in linkedin.com/in/rsaparkh/](https://www.linkedin.com/in/rsaparkh/) | github.com/roma2023

EDUCATION

Carnegie Mellon University

August 2022 – May 2026

Bachelor of Science in Computer Science, Dean's List

Doha, Qatar

- **Relevant Coursework:** Distributed Systems, Software Engineering, Web Application Development, Generative AI, Algorithms & Design, Deep Learning, Machine Learning, Parallel & Sequential Data Structures & Algorithms.
- **Technical Leadership:** Google Developer Student Club Lead, AWS College Contributor Initiative Team Lead.

EXPERIENCE

Software Engineer Intern

June 2025 - August 2025

UBS *Doha, Qatar*

- Built syntifAI: synthetic-data platform (Python/TypeScript/FastAPI) for 100+ table DBs in testing environments.
- Developed schema-aware parsing to auto-discover tables & generate Profile Maps; cut test-data prep time by 85%.
- Optimized pipeline via performance tuning on Linux; throughput +35%, <2s/table using Bash & Git workflows.
- Deploying in UBS Doha; led brainstorming sessions to boost team productivity; scalable for UBS-wide adoption.

AI Engineer Intern

December 2024 - June 2025

Qatar Computing Research Institute (QCRI)

Doha, Qatar

- Built predictive models integrating LLM & RAG for real-time analytics; applied problem solving for consistency.
- Built hybrid CoT-PoT ensembling approach; reduced LLM reasoning tasks computational cost by 95% via planning.
- Integrated open-source frameworks (DeepSeekR1, Qwen2.5); cut costs 80% & collaborated on model decisions.
- Enabled multi-processing & queuing for concurrent execution; critical troubleshooting for workflow efficiency.

Teaching Assistant for Distributed Systems

August 2024 - November 2024

Carnegie Mellon University, CS Department

Doha, Qatar

- Taught via office hours and 1:1 sessions, developing strong verbal communication skills in computer science.
- Applied big data frameworks Ray, MapReduce, Hadoop; used version control & software development lifecycle.
- Mentored 35 students on projects; led brainstorming on system design & performance tuning decision-making.

ML Research Engineer

May 2024 - July 2024

Carnegie Mellon University, Research Department

Doha, Qatar

- Enhanced Pairs Trading by applying NLP into Quant analysis for trading systems; achieved 15% annual return.
- Fine-tuned Llama 3-8B; integrated financial LLMs & lexicons for sentiment analysis; boosted returns by 1.5%.
- Built Python data pipeline using scripting language for 300 tickers; retrieved 300k news samples via Refinitiv.
- Developed custom backtesting platform for trading algorithm optimization; improved development efficiency.

Software Engineer Contributor

March 2023 - May 2024

Amazon Web Services (AWS) - OpenSearch

Seattle, WA, USA

- Contributed to OpenSearch Python & ML libraries; collaborated with teams to enhance stability & efficiency.
- Developed snapshot API support; redesigned test splits & integrated new Python dependency management.
- Featured speaker at OpenSearchCon 2023; shared technology insights for open-source college contributors.

PROJECTS

VibePrint | *Node.js, FastAPI, Tailwind CSS, Three.js, WebGL, TripoSR, AWS*

- Developed platform converting images/text/sketches to 3D models with printing services; self-hosted TripoSR.
- Implemented real-time 3D rendering with file management; added community feed, chat, posts & model sharing.

Lifted AI | *Next.js, FastAPI, LangChain, Supabase, OpenAI*

- Built AI copilot for PMs to accelerate task breakdown, allocation & real-time troubleshooting support systems.
- Implemented embeddings-based risk detection; vector search on Supabase & auto-summarization for meetings.

Find4Cheaper | *Next.js, FastAPI, Supabase, Redis, OpenAI API*

- Built an AI web application aggregating niche marketplace data to compare prices, availability, and shipping tradeoffs.
- Built crawler and real-time search with JSON-LD caching; cut discovery time 90%, indexed 50M+ pages on 500+ sites.
- Launched AI shopping assistant enabling natural language product search with automated ranking and de-duplication.

FileStack – Distributed File System | *Java, C++, Linux, Java RMI, TCP/IP*

- Built distributed file system (Java/C++) on Linux with TCP/UDP networking, replication & fault handling.
- Reduced latency 30% via network routing optimization; collaborative design & performance tuning techniques.

TECHNICAL SKILLS

Programming Languages: Python, Bash, C/C++, Java, JavaScript, TypeScript, Dart, Rust, HTML, CSS, SQL, SML

Frameworks & Tools: Git, React, Node.js, Express, PyTorch, TensorFlow, NumPy, Scikit-learn, Pandas, OpenCV

Technologies: UNIX/Linux, Docker, AWS, Azure, Vercel, LangChain, Redis, Firebase, TCP/IP, UDP, SQL Databases